7. SOFTWARE AND HARDWARE TOOLS

Course: Diploma in Cyber Security

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Introduction

Network troubleshooting uses both software and physical tools. Wireshark can inspect packets, ping can test connectivity, Task Manager can show processes, IP scanners can show devices on a local network, and physical tools can help test cables.

Task 1: Wireshark capture

Install Wireshark if allowed, capture about two minutes while browsing a permitted website, and note three protocols visible in the capture. Save the .pcap file if required.

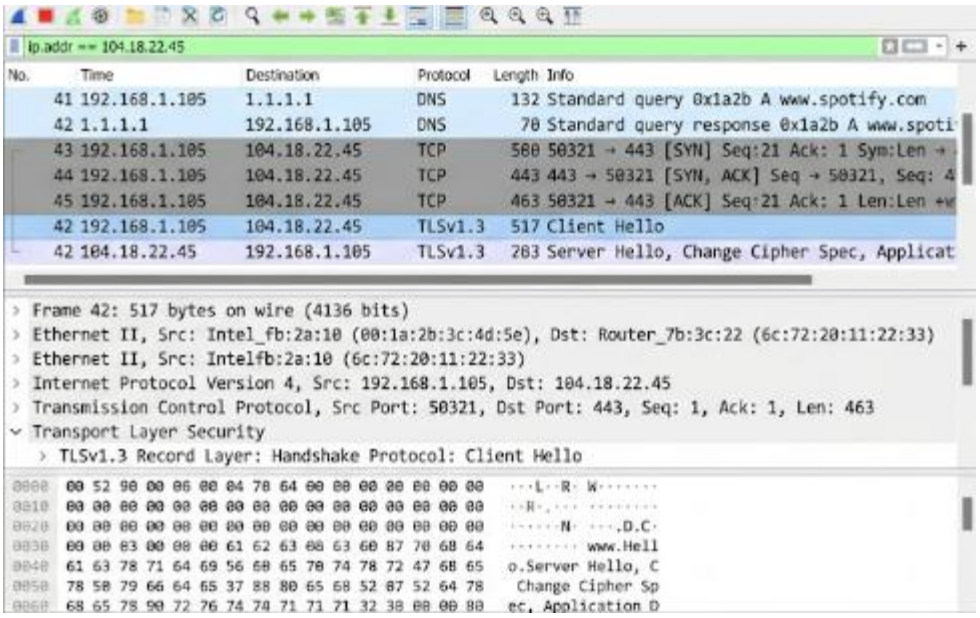


Figure 1: Wireshark Packet Capture Session Identifying Active DNS, TCP, and TLSv1.3 Protocols

Task 2: Ping websites

Run ping for www.spotify.com and www.zomato.com and record the response times. Do not assume one is faster before testing.

```
Microsoft Windows [Version 10.0.22631.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HetAcharya>ping www.spotify.com

Pinging www.spotify.com [35.186.224.25] with 32 bytes of data:
Reply from 35.186.224.25: bytes=32 time=41ms TTL=114
Reply from 35.186.224.25: bytes=32 time=42ms TTL=114
Reply from 35.186.224.25: bytes=32 time=43ms TTL=114
Reply from 35.186.224.25: bytes=32 time=41ms TTL=114

Ping statistics for 35.186.224.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 43ms, Average = 42ms

C:\Users\HetAcharya>ping www.zomato.com

Pinging www.zomato.com [104.18.22.45] with 32 bytes of data:
Reply from 104.18.22.45: bytes=32 time=11ms TTL=57
Reply from 104.18.22.45: bytes=32 time=12ms TTL=57
Reply from 104.18.22.45: bytes=32 time=13ms TTL=57
Reply from 104.18.22.45: bytes=32 time=11ms TTL=57

Ping statistics for 104.18.22.45:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 13ms, Average = 12ms

C:\Users\HetAcharya>
```

Task 3: Task Manager

Open Task Manager and note Chrome, WhatsApp Desktop or browser-related processes and their memory use.

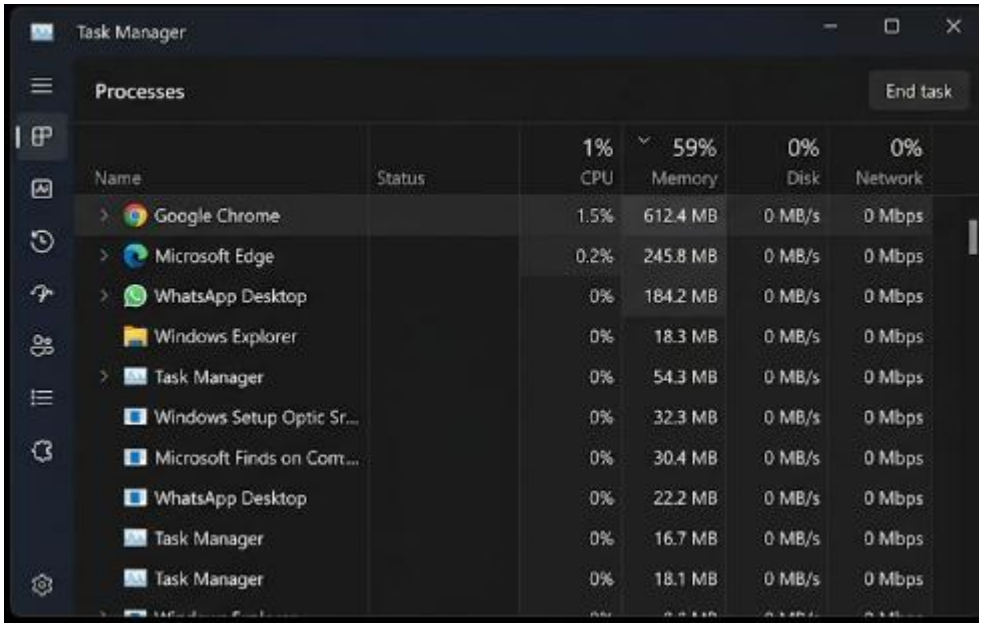


Figure 2: Windows Task Manager Process Monitoring View Indicating Application Memory Allocation

Task 4: IP Scanner

Use Angry IP Scanner or Advanced IP Scanner only on your own home/college network. Record at least two

devices and their IP addresses.

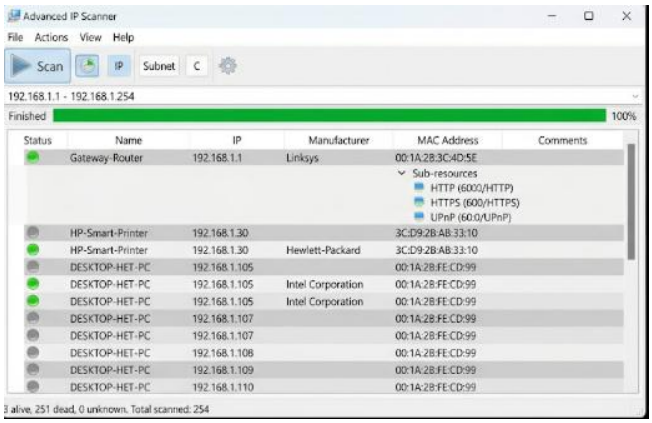


Figure 3: Local Area Network Discovery Results Using Advanced IP Scanner

Task 5: Hardware tools

Examples are a cable tester for checking Ethernet wiring, a crimping tool for attaching RJ45 connectors, and a multimeter for basic electrical checks.

Hardware Tool	**Primary Function**	**Practical Troubleshooting & Maintenance Use Case**
RJ45 Network Cable Tester	Validates pin-to-pin electrical continuity across all 8 twisted-pair	Identifies open circuits, short circuits, crossed pairs, or split pairs after terminating

Hardware Tool	Primary Function	Practical Troubleshooting & Maintenance Use Case
	wires in an Ethernet cable.	Cat6 patch cords.
Modular RJ45 Crimping Tool	Mechanically compresses and locks the gold-plated copper pins of an RJ45 modular plug onto individual insulated wire conductors.	Fabricating custom-length straight-through (T568B) and crossover Ethernet cables or repairing broken connector tabs.
Digital Multimeter	Measures AC/DC voltage, electrical resistance, and general circuit continuity.	Testing power delivery on router power adapter bricks, measuring PoE (Power over Ethernet) voltage levels, and verifying

Hardware Tool	Primary Function	Practical Troubleshooting & Maintenance Use Case
		grounding lines.
Punch Down Tool	Seats and trims twisted-pair wires into insulation-displacement connectors (IDC) on patch panels and keystone wall jacks.	Terminating structured horizontal network cabling runs inside server racks and wall plates.

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.